



Center for Operational Oceanographic Products and Services
NATIONAL OCEAN SERVICE

Comparing Flood Indicators Using Stationary and Nonstationary Extreme Water Level Analysis Across U.S. Coastal Regions

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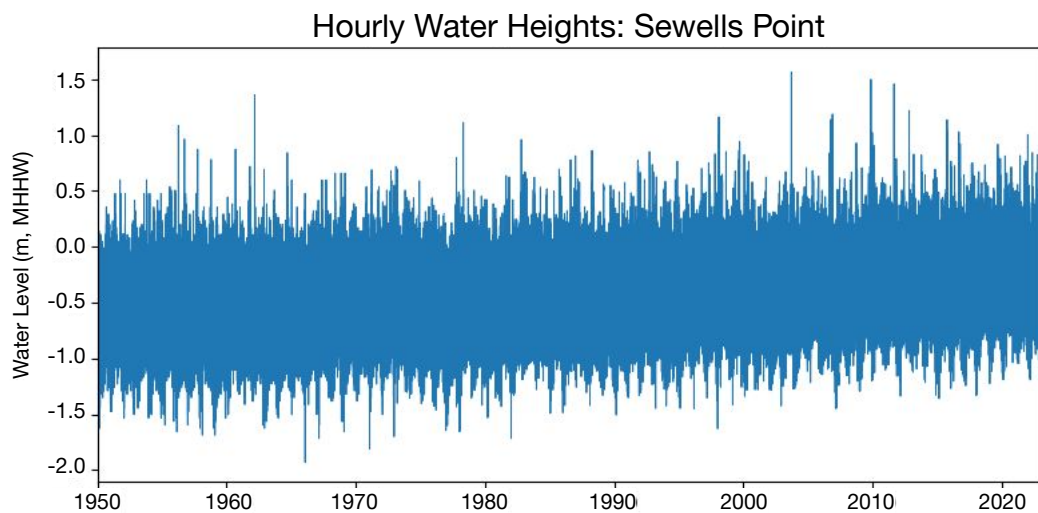
Why Coastal Flooding Matters

- 129 million people (~40% of U.S. population) live on the coast, where extreme water levels pose a growing flood risk (NOAA OCM)
- Sea level rise increases flooding risk by raising water levels
- Flood events disrupt communities, damage infrastructure, and degrade ecosystems
 - Nearly \$7 billion in damages in 2023 (NOAA OCM)



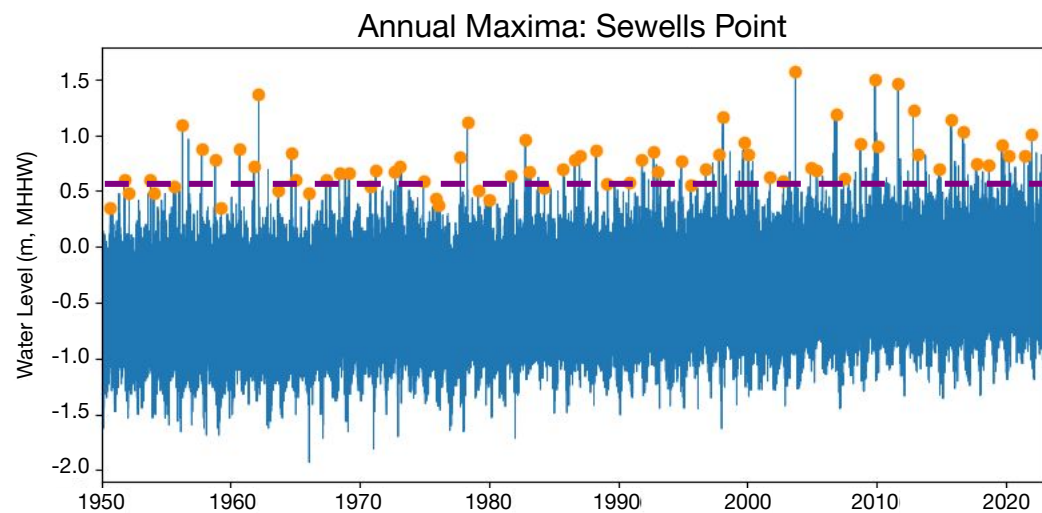
Measuring Water Levels

- Tide gauges have continuously measured changes in water level along the U.S. coast
 - Over 170 years, 200 stations
- Long-term records and real-time data support tide predictions, storm surge modeling, flood planning
 - Ports, protecting infrastructure
- Water level patterns differ across locations

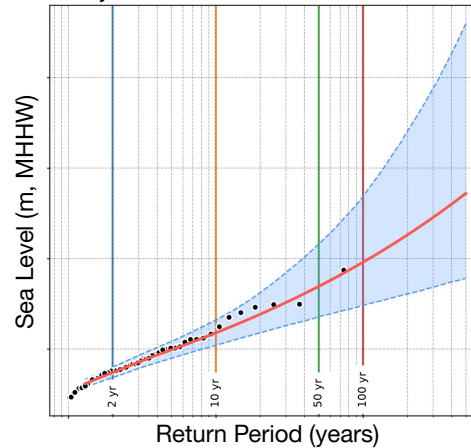


Extreme Water Levels

- **Extreme Water Level (EWL):**
observation with exceptionally high water height
 - Definition can vary
- Result from storm surge, tsunamis, high tides, sea level rise, etc.
- EWL Products: return level curves, rankings of top events
- Help estimate height and frequency of potentially damaging floods
 - Coastal planning, infrastructure design, hazard mitigation



Stationary GEV Return Level Curve: Sewells Point

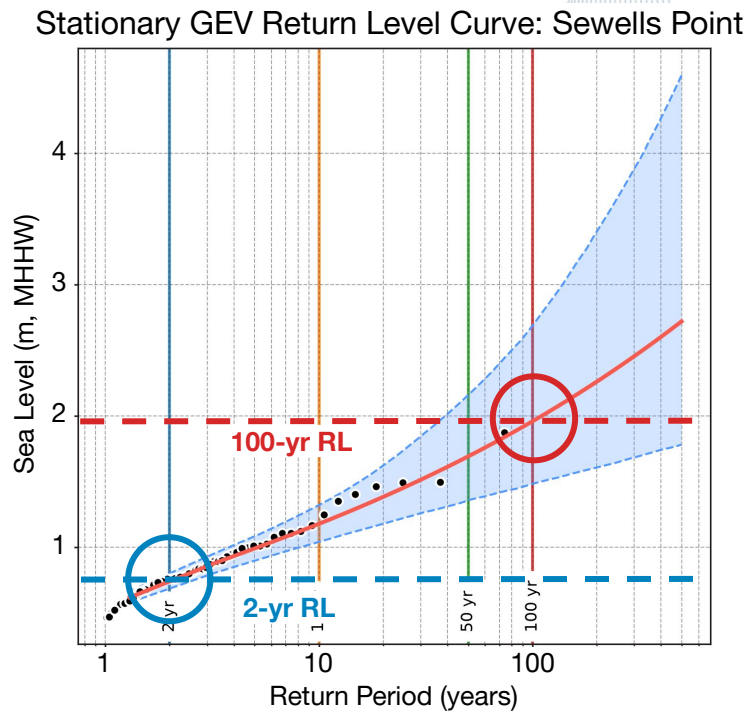


Top 3 Events for Sewells Point

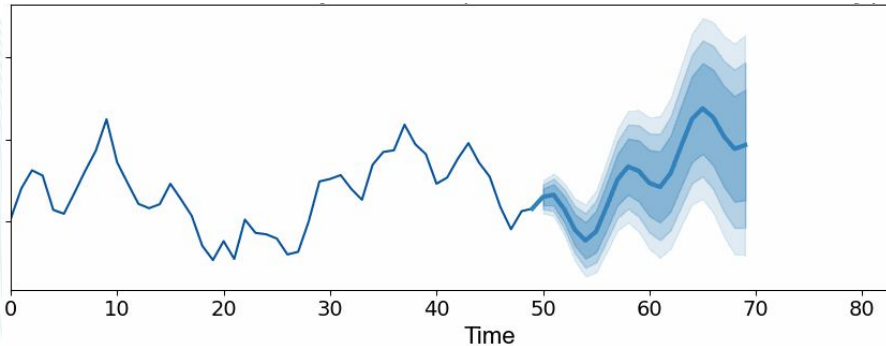
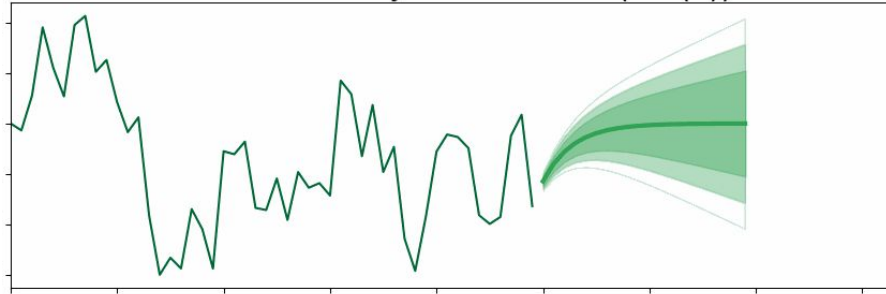
Rank	Event	WL (m)
1	8/23/1993 Hurricane Emily	1.604
2	9/18/2003 Hurricane Isabel	1.564
3	11/12/2009 Hurricane Ida	1.499

Extreme Value Analysis

- **Extreme Value Analysis:** statistical method used to model and quantify the probability of EWLs
 - Generalized Extreme Value (GEV) distribution
 - **Return level:** water height associated with a certain return period
 - **Return period:** average length of time between exceedances of that water level
- A 100-year return level has a 1% chance of being exceeded in any given year
 - 2-year return level: 50% chance



Stationary & Nonstationary Analysis



GIF Credit: Kusum Khatri

Stationary GEV Model

- Traditionally-used approach
- Statistical distribution of EWLs remain stationary through time

Nonstationary GEV Model

- Sea level rise, climate cycles, can change patterns over time
- Statistical distribution of EWLs varies with time

Project Objectives

Stationary vs. Nonstationary EWLs



Do stationary and nonstationary extreme value models differ in their estimates of coastal flood return levels?

Primary Drivers of Nonstationarity



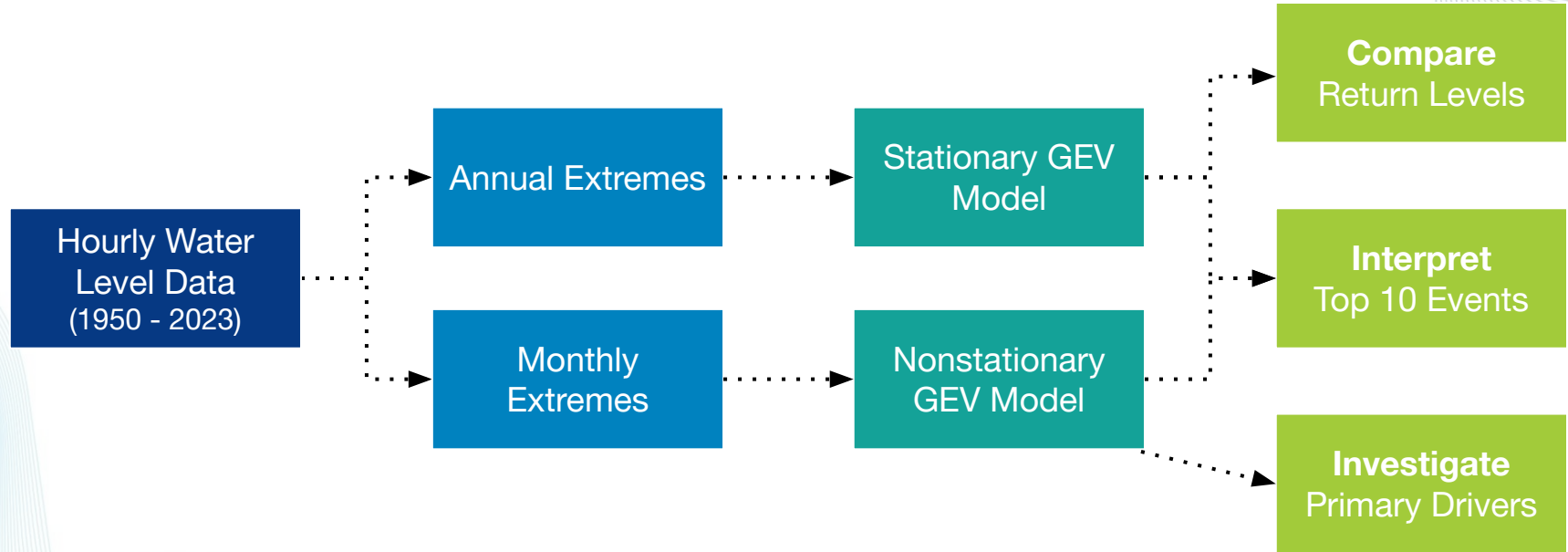
What are the primary drivers of nonstationarity in EWLs, and do their relative contributions vary across U.S. coastal regions?

Historical Interpretation of Top 10 Events

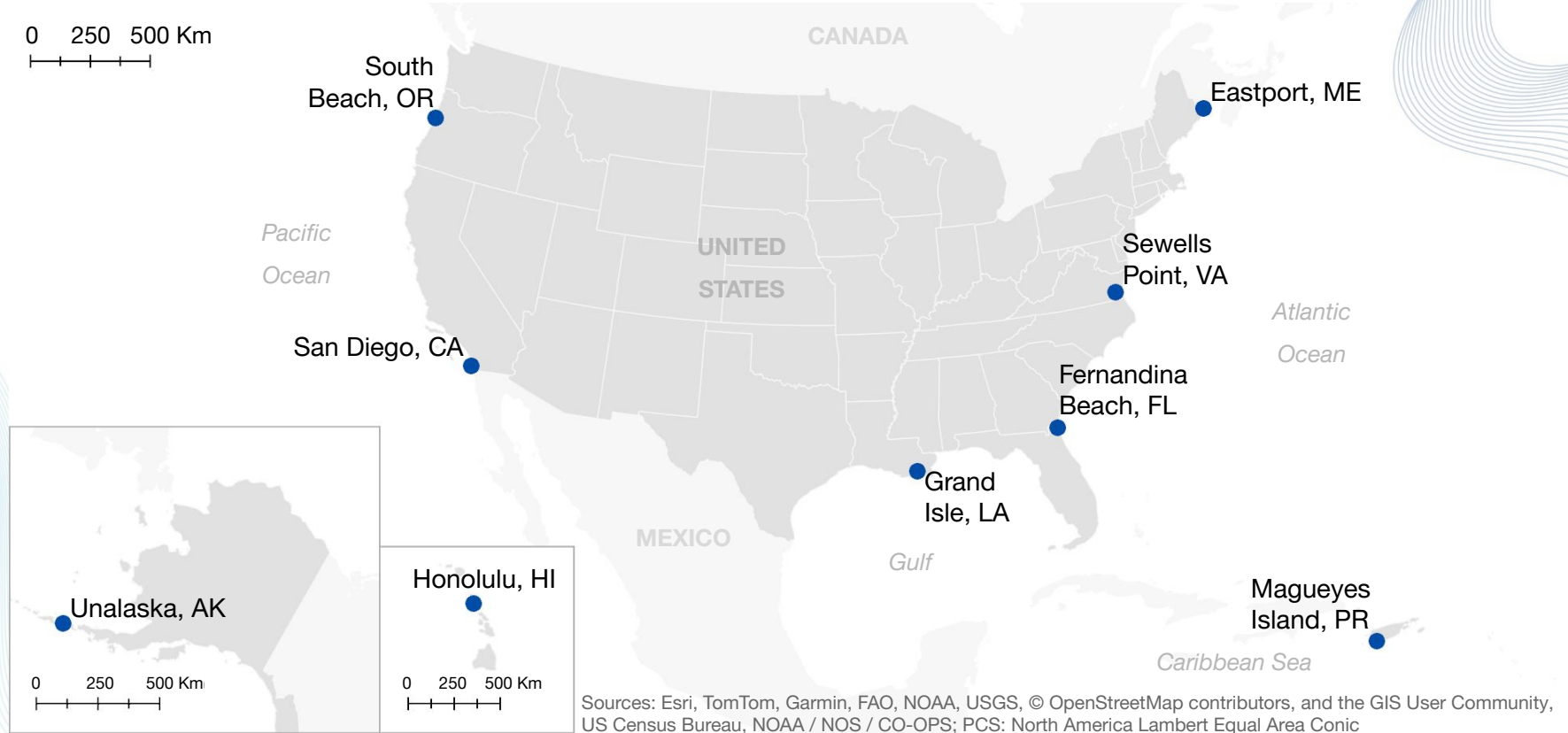


Does a nonstationary framework change the historical interpretation of the ten highest observed water levels at each station?

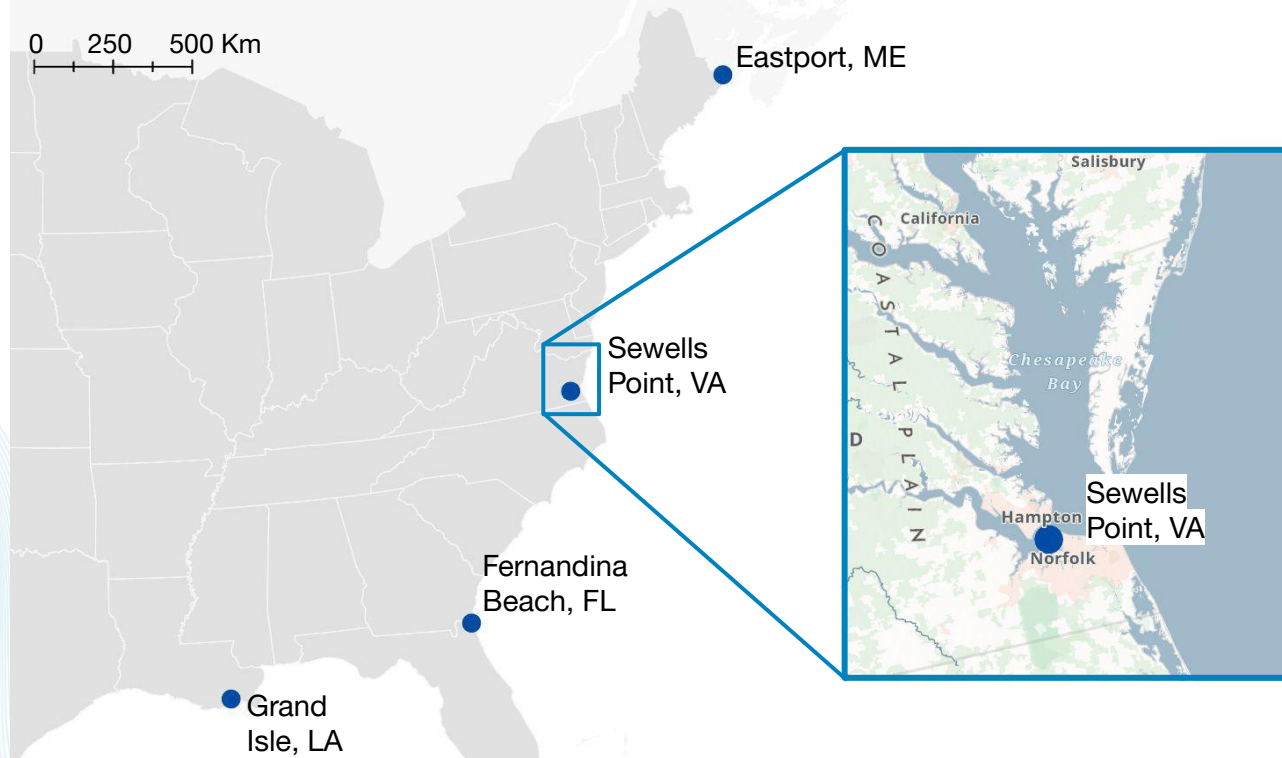
Methods: Overview



Methods: Representative Stations



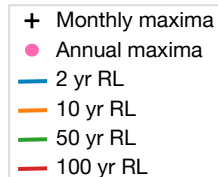
Methods: Representative Stations



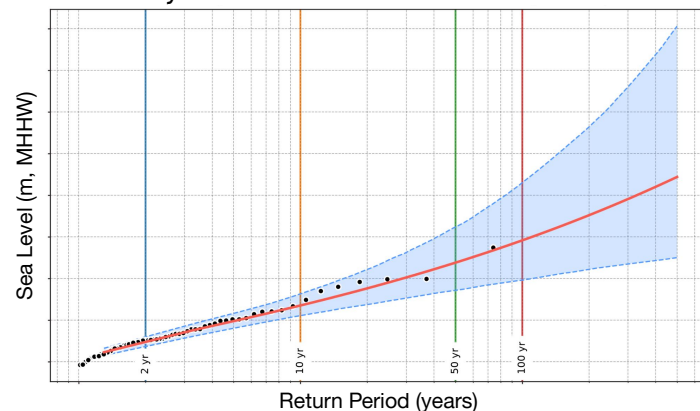
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community, US Census Bureau, NOAA / NOS / CO-OPS
PCS: North America Lambert Equal Area Conic

Methods: Stationary GEV Analysis

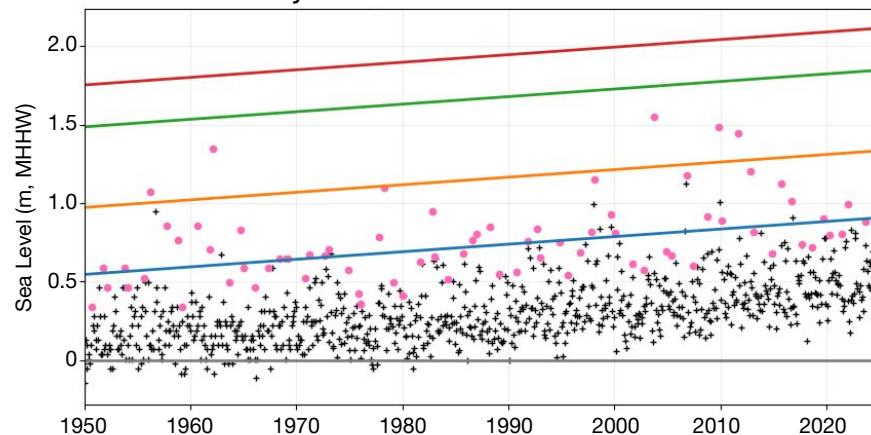
- Annual extremes
- 3 parameters:
 - **Location:** central level of extremes
 - **Scale:** variability of the extremes
 - **Shape:** behavior of the upper tail
- Remain stationary through time
- Referenced to the midpoint of the 1983-2001 National Tidal Datum Epoch (1992)



Stationary GEV Return Level Curve: Sewells Point

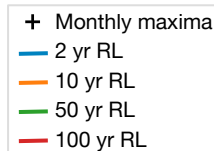


Stationary GEV Return Levels: Sewells Point

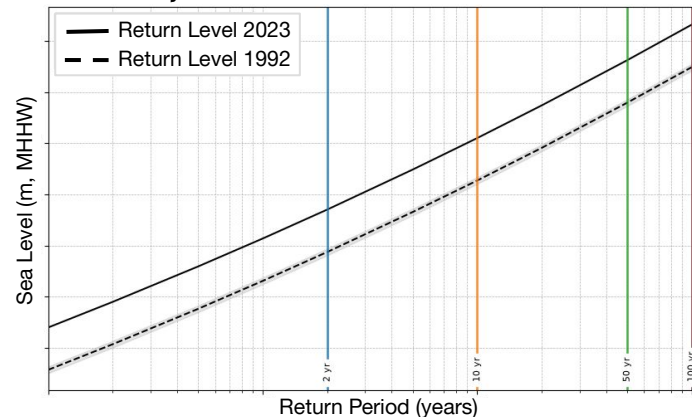


Methods: Nonstationary GEV Analysis

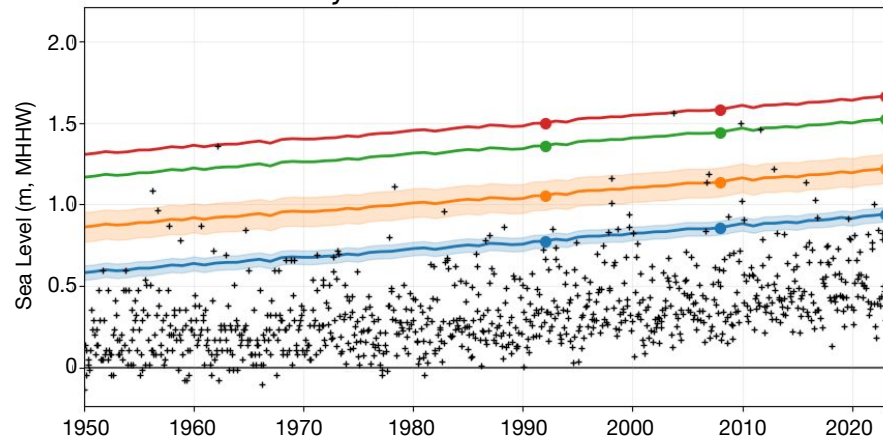
- Monthly extremes
- Iterative model selection process
 - Model tests different combinations of time-varying components
- Up to 14 parameters
- Produces time-varying return level curves



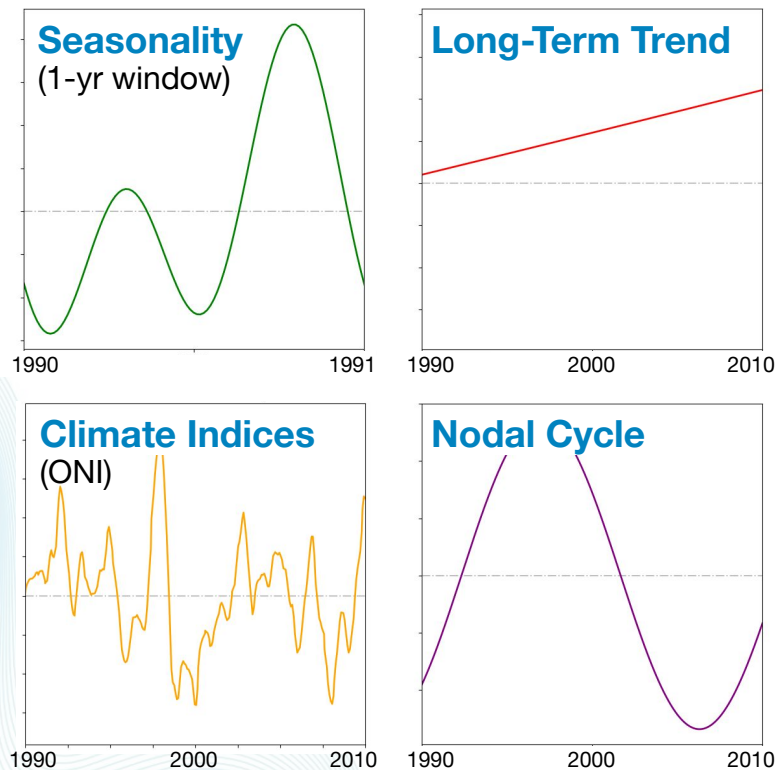
Nonstationary GEV Return Level Curve: Sewells Point



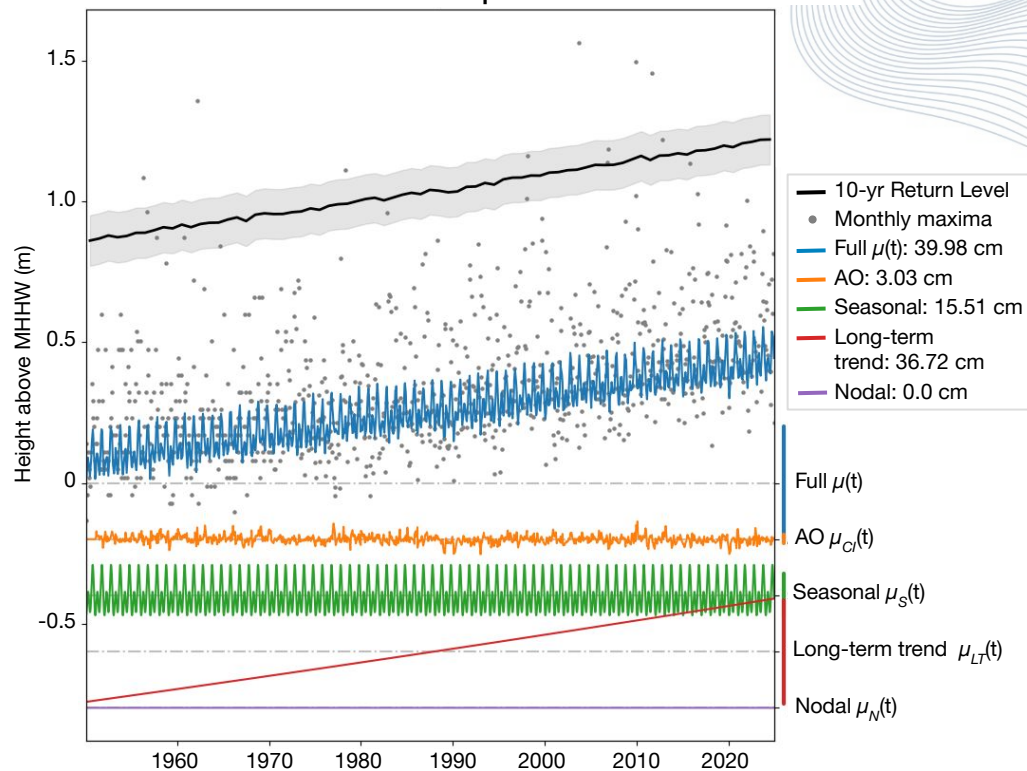
Nonstationary GEV Return Levels: Sewells Point



Methods: Drivers of Nonstationarity



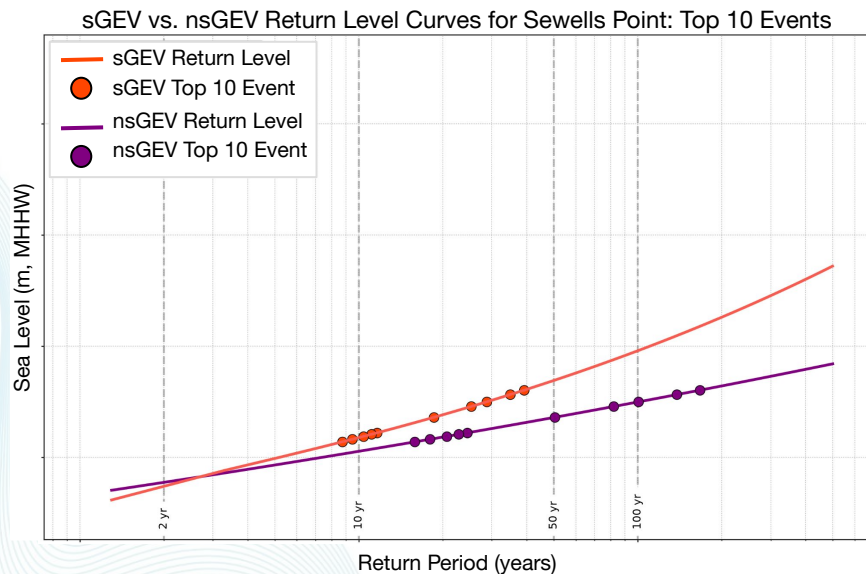
Location Parameter Components: Sewells Point



Methods: Top 10 Events

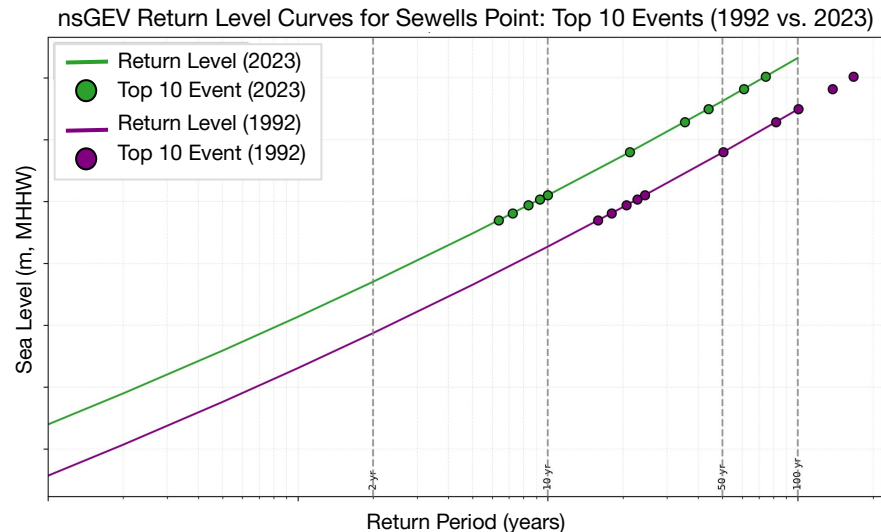
Stationary vs. Nonstationary

Model Comparison



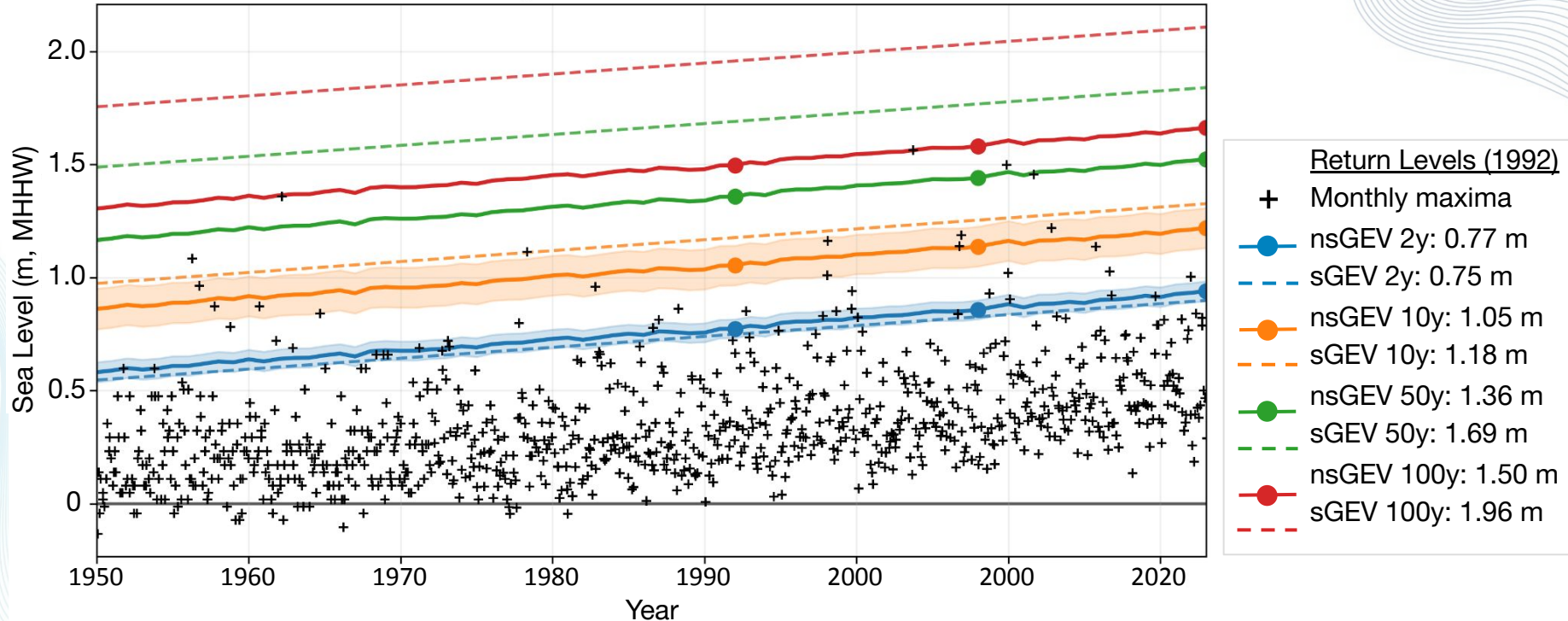
Nonstationary: 1992 vs. 2023

Time Comparison



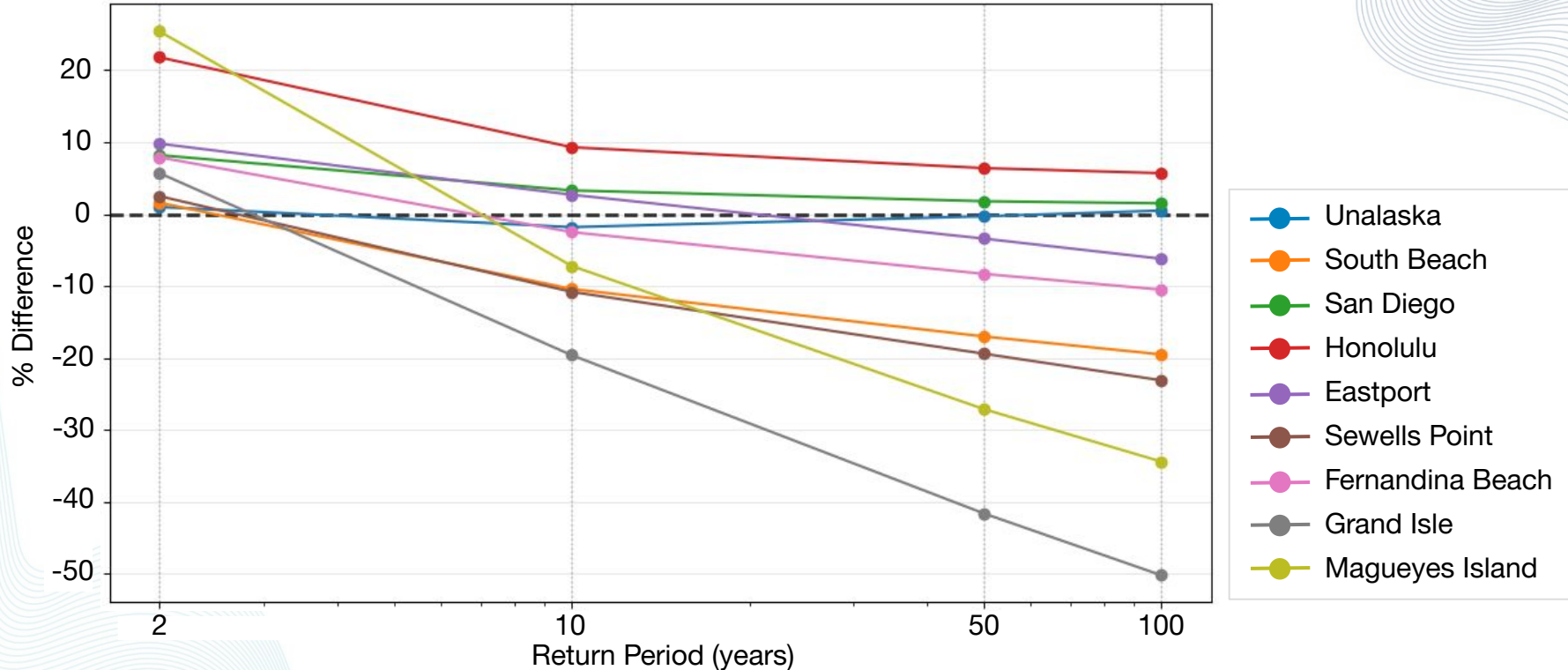
Results: Stationary vs. Nonstationary Return Levels

Nonstationary vs. Stationary GEV Return Levels: Sewells Point

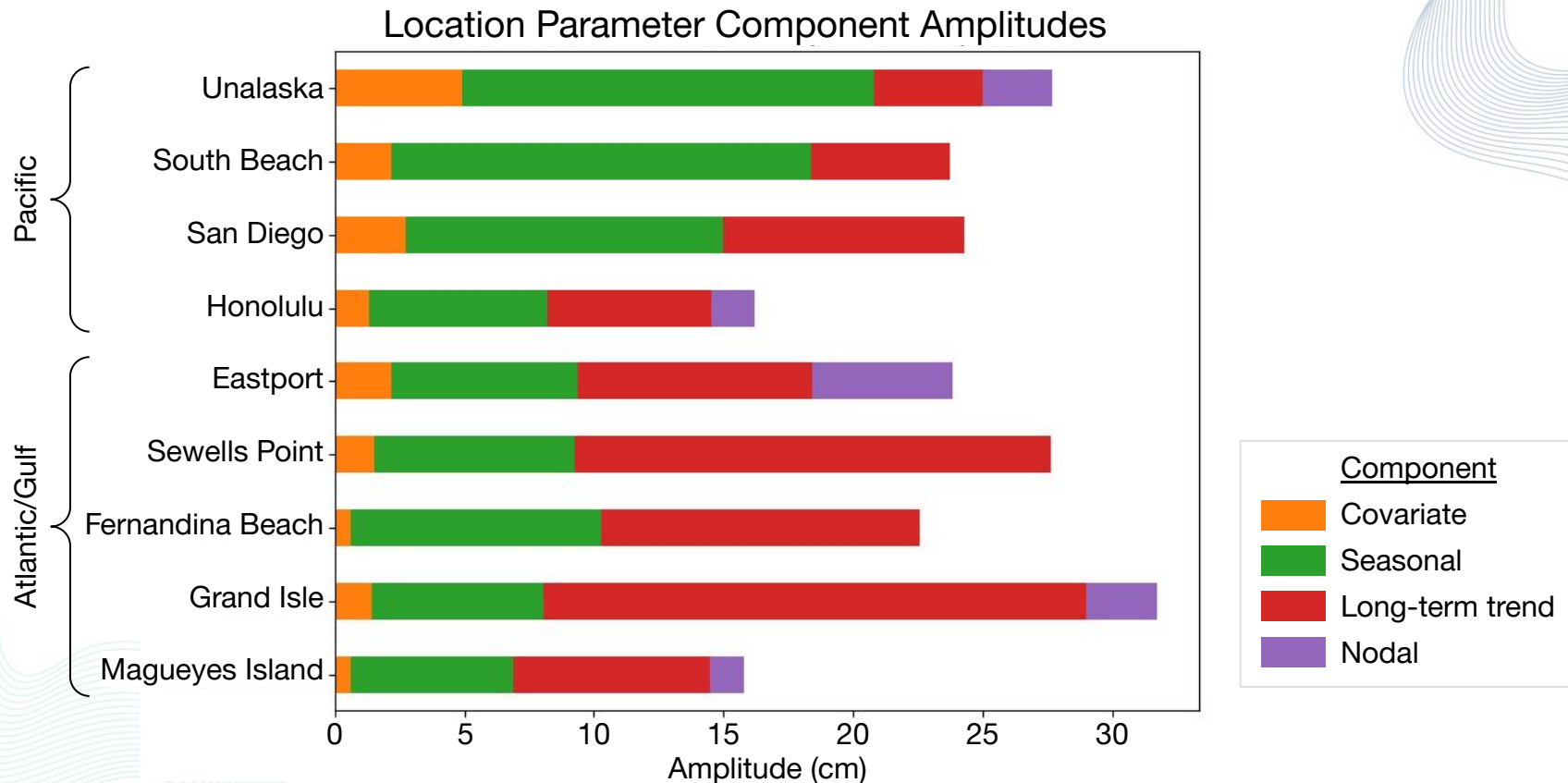


Results: Stationary vs. Nonstationary Return Levels

Percent Difference in Nonstationary Return Levels Relative to Stationary

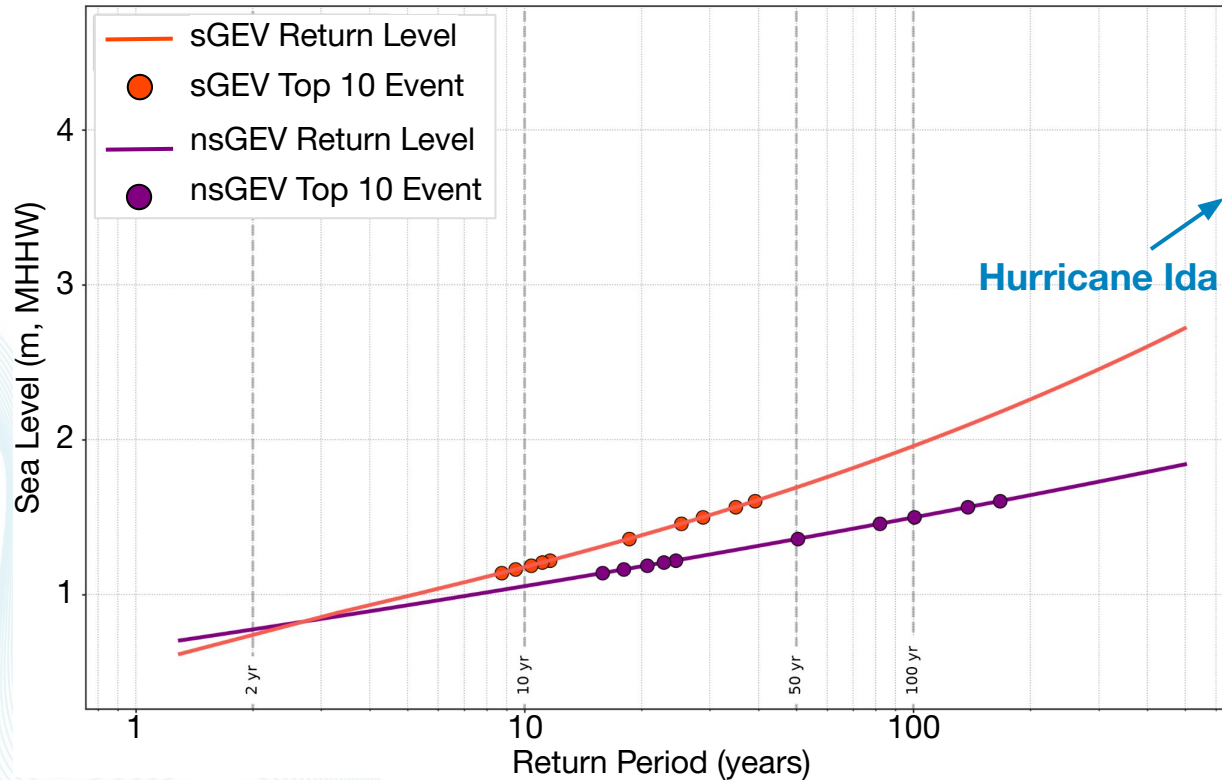


Results: Drivers of Nonstationarity



Results: Top 10 Events

Stationary vs. Nonstationary GEV Return Level Curves for Sewells Point: Top 10 Events

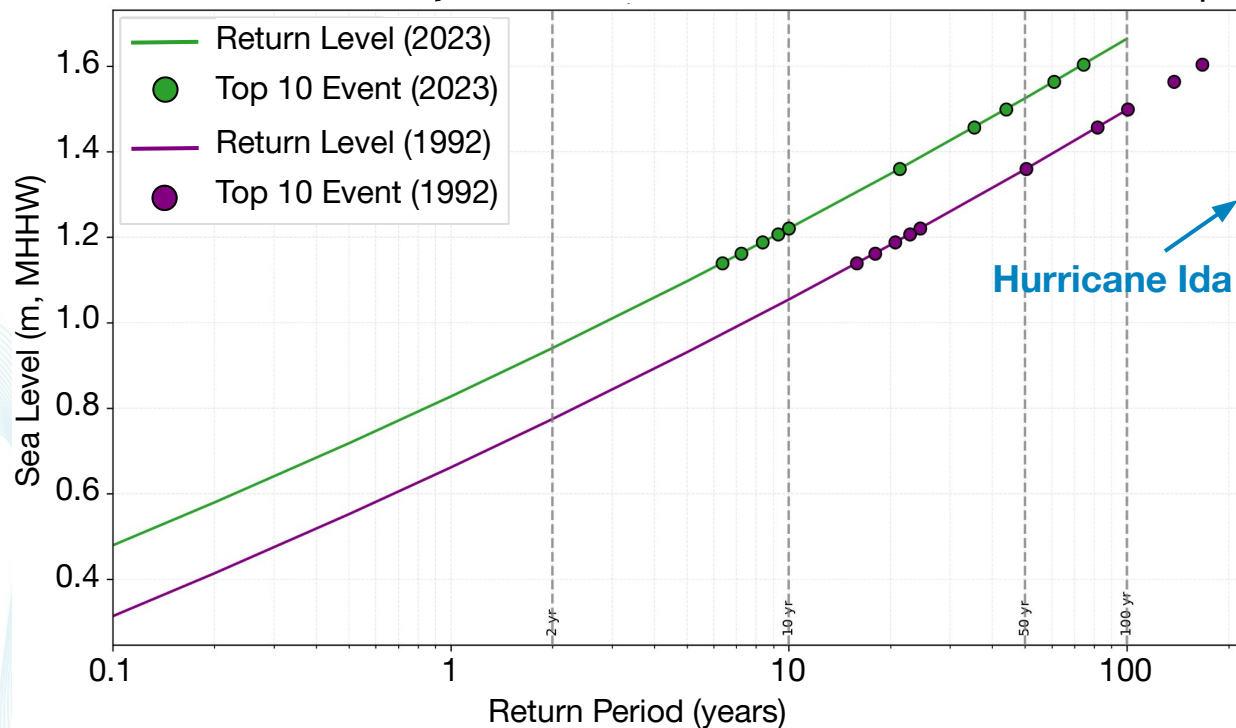


Rank	Event	Water Level (m)	sGEV RP (yr)	nsGEV RP (yr)
1	8/23/1933	1.604	39.1	167
2	9/18/2003	1.564	34.9	137.9
3	11/12/2009	1.499	28.7	100.7
4	8/28/2011	1.457	25.3	81.9
5	3/7/1962	1.36	18.6	50.4
6	10/29/2012	1.22	11.6	24.5
7	9/18/1936	1.207	11.1	22.8
8	11/22/2006	1.188	10.4	20.6
9	2/5/1998	1.162	9.5	18
10	10/7/2006	1.139	8.7	15.9

- Most stations followed similar pattern, except San Diego and Honolulu

Results: Top 10 Events

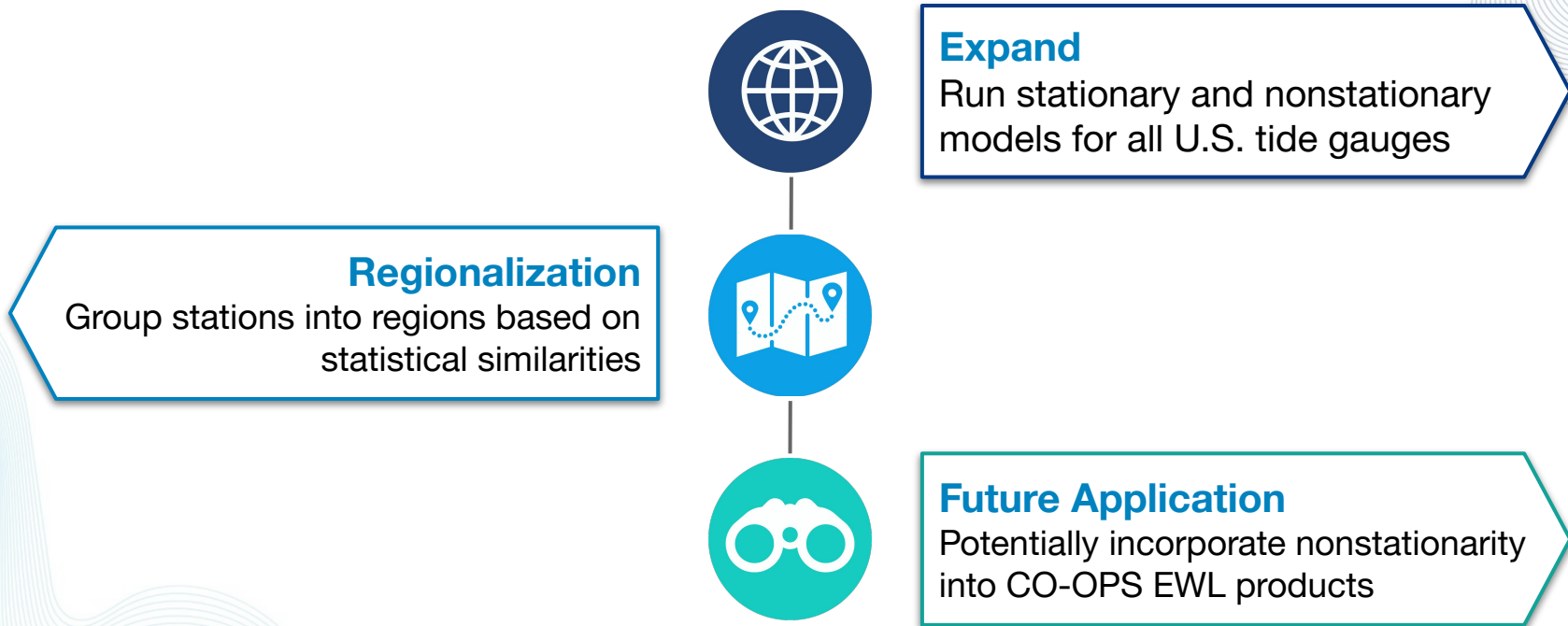
Nonstationary Return Level Curves for Sewells Point: Top 10 Events (1992 vs. 2023)



Rank	Event	Water Level (m)	1992 RP (yr)	2023 RP (yr)
1	8/23/1933	1.604	167	74.4
2	9/18/2003	1.564	137.9	60.9
3	11/12/2009	1.499	100.7	43.9
4	8/28/2011	1.457	81.9	35.4
5	3/7/1962	1.36	50.4	21.3
6	10/29/2012	1.22	24.5	10
7	9/18/1936	1.207	22.8	9.3
8	11/22/2006	1.188	20.6	8.4
9	2/5/1998	1.162	18	7.2
10	10/7/2006	1.139	15.9	6.4

- All stations with positive long-term trends followed similar pattern

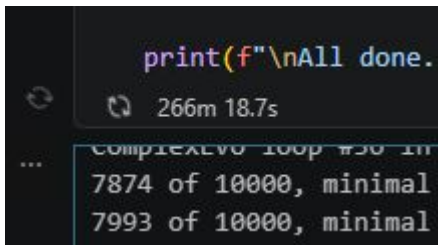
Future Work



Barriers & Personal Accomplishments

Barriers

- Initial delays in setting up
- Long model runtimes



```
print(f"\nAll done.\n")
266m 18.7s
...
complex loop 10000 in
7874 of 10000, minimal
7993 of 10000, minimal
```

Personal Accomplishments

- Strengthened Python skills
- Gained understanding of coastal oceanography and water level data
 - Visited a tide gauge
- Learned about NOAA
 - Attended conferences, visited NOAA Cooperative Oxford lab

Conclusions

1

Accounting for time-varying processes changed both return level estimates and the interpretation of past extreme events. Nonstationarity may be important for understanding how flood risk evolves over time.

2

The effects of nonstationarity were not consistent across stations or return periods, indicating that changes in EWLs may be location specific.

3

The apparent differences in nonstationary drivers and regional patterns are preliminary; analysis across more stations is needed to determine whether station-level differences form consistent regional patterns.

Thank you!

Acknowledgements

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